

Data Science 6

Mark

Lecture 3

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Data, Variables and Arrays

1 Working with Data

Consider social implications as data scientist, power that comes with it to change. Yet also social implications can lead to pre-conceived notions which change and restrict the model off the get go.

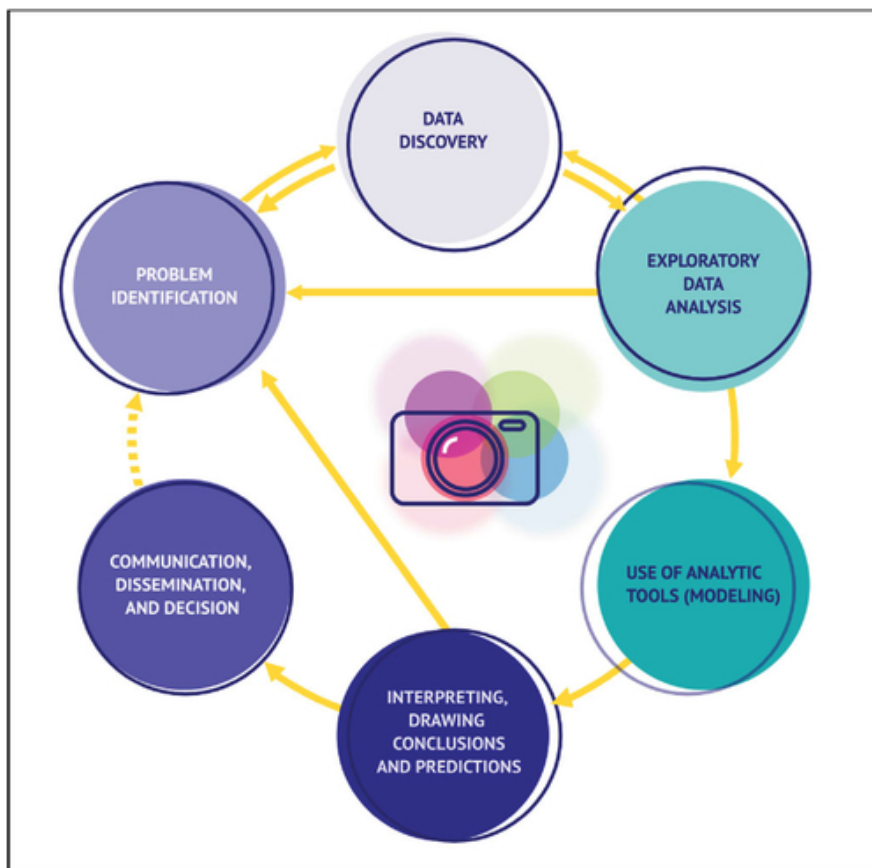


Figure 1: The life cycle of data science.

2 Variables

A **dataset** is a bunch of data gathered together. **tabular data** is just data with columns and rows, basically a table. **Domain** is the set of possible values of a variable. **Operationalization**: going from abstract data to having a variable and way to measure.

Ex: for a state population var domain is $\mathbb{Z}^+$.

Note. In python some people call python names variables but in this class variables differ from python names.

3 Variable Types

Recall variable types from AP Stats. Numerical vs Categorical. Within numerical we have discrete and continuous variables. Within categorical we have ordinal and nominal. Ordinal is simply categories with inherent ordering.

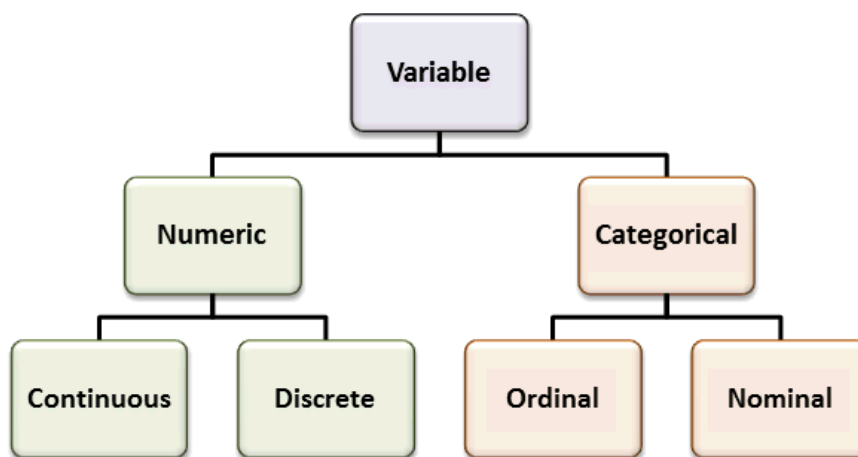


Figure 2: Diagram of var types

Ex: Area codes are categorical since you **can't do arithmetic with it**.

4 Units of Analysis

Units of analysis is linked to the concept of vars because vars describe certain types of entities. Some examples are Individuals, groups like families and variables to understand that could be like income or num of people.

5 Arrays

Arrays are simply a data type where the name leads to a list of data types you can store in it. "A sequential collection of values". Remember they are zero indexed, and same type in all array. Python arithmetic are all element-wise operations.

6 Numpy